

AI/ML Engineering Roadmap

A Complete Self-Study Path from Foundations to Professional AI Engineering

Study Order

Phase 0	Math Foundations + Python (run in parallel throughout)
Phase 1	Python · Statistics · EDA · Visualization (GFG Weeks 0–4)
Phase 2	SQL + Databases
Phase 3	Machine Learning + WQU Data Science Projects (GFG Weeks 5–10)
Phase 4	Git · Docker · CI/CD · Core MLOps
Phase 5	Deep Learning+ WQU DL Projects + Reinforcement Learning + DSA (GFG Weeks 11–16)
Phase 6	Modern AI Engineering (NLP · CV · LLMs · GenAI · RAG · Vector DBs · Agents)
Phase 7	Advanced Track — Cloud · Data Engineering · Google Professional ML Engineer

PHASE 1 Python · Statistics · EDA · Visualization

✓ Fully covered by GFG Weeks 0–4. No external course required.

- 1.1 Data Science Overview — *GFG Week 0*
 - 1.2 Python Foundations — *GFG Week 1*
 - 1.3 Statistics for Data Science — *GFG Week 2*
 - 1.4 EDA with Pandas & NumPy — *GFG Week 3*
 - 1.5 Data Visualization — Matplotlib, Seaborn, Plotly — *GFG Week 4*
- GFG Course: <https://www.geeksforgeeks.org/batch/ds-16?tab=Resources>

PHASE 0 Python + Math Foundations

Run Python (0.0) alongside every phase. Complete Math (0.1–0.4) before starting Phase 1.

0.0 — Python

- **Python Programming MOOC 2026** — *University of Helsinki · free · world-class*
<https://programming-26.mooc.fi/>

0.1 — Calculus

- **AP Calculus AB** — *Khan Academy*
<https://www.khanacademy.org/math/ap-calculus-ab>

0.2 — Linear Algebra

- **Linear Algebra** — *Khan Academy*
<https://www.khanacademy.org/math/linear-algebra>

0.3 — Statistics & Probability

- **Statistics & Probability** — *Khan Academy*

<https://www.khanacademy.org/math/statistics-probability>

Alternative Path — Mathematics for Machine Learning (if Khan Academy feels too disconnected from AI)

Mathematics for Machine Learning Specialisation — Imperial College London · Coursera · audit free · Linear Algebra, Multivariate Calculus & PCA

0.1 — Calculus

- **Multivariate Calculus** — Imperial College London
<https://www.youtube.com/watch?v=cWZLPv4ZJhE&list=PLiiljHvN6z193BBzS0Ln8NnqQmzimTW23>
- **Essence of Calculus** — 3Blue1Brown · watch first
<https://www.youtube.com/playlist?list=PLZHQObOWTQDMSr9K-rj53DwVRMYO3t5Yr>

0.2 — Linear Algebra

- **Linear Algebra for Neural Networks** — Imperial College London
https://www.youtube.com/playlist?list=PLiiljHvN6z1_o1ztXTKWPrShrMrBLo5P3
- **Essence of Linear Algebra** — 3Blue1Brown · must watch
https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

— PCA (after linear algebra)

- **PCA Course** — Marc Peter Deisenroth · co-author of Mathematics for Machine Learning
https://www.youtube.com/playlist?list=PL93aLKqThq4iDbY_wxfh3WqxkkzqH4aoo
- **Companion Textbook (free PDF)** — mml-book.com
<https://mml-book.com>

⚠ GFG does not teach SQL. Complete the resources below before Phase 3.

2.1 SQL Murder Mystery — *Joon Park and Cathy He*
<https://mystery.knightlab.com/>

2.2 SQL Tutorial 1 — *Mode Analytics · primary*
<https://mode.com/sql-tutorial/>

2.3 SQL Tutorial 2 — *SQL Zoo*
https://sqlzoo.net/wiki/SQL_Tutorial

2.4 PostgreSQL Basics — *YouTube · optional but recommended*
<https://www.youtube.com/watch?v=qusjTv8N1cA>

PHASE 3 Machine Learning + WQU DS Projects

✅ Fully covered by GFG Weeks 5–10. No external course required.

3.1 ML Basics + Regression — *GFG Week 5*

3.2 Model Evaluation + SHAP / Model Explainability + Classification — *GFG Week 6 + SHAP*

Do SHAP / Model Explainability immediately after Day 36. It takes one afternoon and you'll use it in every project after.

3.2.1 SHAP / Model Explainability — <https://shap.readthedocs.io/en/latest/>

3.3 ML Algorithms — Decision Trees, SVM, KNN, Naive Bayes — *GFG Week 7*

3.4 Ensemble Learning — XGBoost, LightGBM, CatBoost — *GFG Week 8*

3.5 Unsupervised Learning — Clustering, GMM, DBSCAN — *GFG Week 9*

3.6 Dimensionality Reduction + Deployment Introduction — *GFG Week 10*

✓ **Where it fits:** Complete after GFG Weeks 5–10. Watch CS229 as a companion or immediately after to solidify the math behind everything GFG taught you.

Stanford University · Full Graduate Course · 27+ hours · Free on YouTube 

<https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShMGgzqpVfbU>

This is the one. Andrew Ng's full Stanford graduate-level ML course — not the Coursera version. This is the original, with full mathematical derivations. Every algorithm you touched in GFG (regression, SVMs, decision trees, EM, RL) is explained from first principles here. Lecture notes alone are some of the best ML reading material ever written.

What you will learn:

- Linear/logistic regression with full mathematical derivation
- Generative vs discriminative models — GDA, Naive Bayes
- SVMs, kernels, and the full dual formulation
- Decision trees, ensemble methods, boosting
- EM algorithm and mixture of Gaussians
- PCA, ICA, factor analysis
- Reinforcement Learning — MDPs, value iteration, policy gradient
- Debugging ML models and error analysis (lecture 13 alone is worth everything)

3 c — WQU Data Science Projects

→ **Applied Data Science Lab**

<https://www.wqu.edu/adsl>

PHASE 4 Git · Docker · CI/CD · Core MLOps

⚠ GFG does not teach these tools. Complete before moving to Deep Learning.

4.1 — Git & GitHub

- **Git & GitHub Full Course** — *freeCodeCamp*
<https://www.freecodecamp.org/news/git-and-github-crash-course-for-beginners/>
- **Microsoft Learn for GitHub** — *Microsoft*
<https://learn.microsoft.com/en-us/training/github/>

4.2 — Docker

- **Docker Tutorial for Beginners** — *TechWorld with Nana · YouTube*
<https://www.youtube.com/watch?v=3c-iBn73dDE>
- **Docker for Beginners** — *freeCodeCamp*
<https://www.freecodecamp.org/news/an-introduction-to-docker-and-containers-for-beginners/>

4.3 — CI/CD

- **GitHub Actions Full Course** — *YouTube*
<https://www.youtube.com/watch?v=mFFXuXjVgkU>

4.4 — Core MLOps

- **Made With ML — MLOps** — *madewithml.com · best free MLOps resource available · primary*
<https://madewithml.com>
- **MLOps Zoomcamp** — *DataTalksClub · alternative / highly practical*
<https://datatalks.club/blog/mlops-zoomcamp.html>

PHASE 5 Deep Learning + WQU DL Projects + Reinforcement Learning + DSA

✓ Fully covered by GFG Weeks 11–16. No external course required.

5.1 Introduction to Deep Learning — *GFG Week 11*

5.2 Optimisation + MLP + Backpropagation — *GFG Week 12*

5.3 CNN + RNN + Time Series — *GFG Week 13*

5.4 LSTM, GRU, Transformers — *GFG Week 14*

5.5 GANs + Autoencoders — *GFG Week 15*

✓ GFG backup

5.6 Practical Deep Learning *fast.ai*

<https://course.fast.ai/>

5.5 Dive into Deep Learning

<https://d2l.ai/>

5.2 Andrej Karpathy: Neural Networks — Zero to Hero

Where it fits: Add these alongside or immediately after GFG Weeks 11–16. MIT 6.S191 is your reinforcement layer. Karpathy is your deep understanding layer.

Former Tesla AI Director, ex-OpenAI · Full YouTube Series · Free 🔗

<https://www.youtube.com/playlist?list=PLAqhlrjkxbuWI23v9cThsA9GvCAUhRvKZ>

This is the most important deep learning resource on the internet right now. Karpathy builds everything from absolute scratch — backpropagation, micrograd, character-level language models, transformers, GPT — line by line in Python. No black boxes. By the end you have built GPT yourself and understand every line of it.

What you will learn:

- Backpropagation from scratch — building micrograd (your own autograd engine)

- makemore — building a character-level language model from scratch
- MLPs, batch normalisation, weight initialisation — built from scratch
- WaveNet-style architectures
- "Let's build GPT from scratch" — the single most important video
- Tokenisation deep dive
- "Let's reproduce GPT-2" — building a 124M parameter GPT-2 from scratch

Why this is non-negotiable: Karpathy has the rare ability to make complex things genuinely simple. His "Let's build GPT" video alone has been called the best educational content on AI ever made. This is what separates engineers who use LLMs from engineers who understand them

5.3 #1 — MIT 6.S191: Introduction to Deep Learning (Alexander Amini)

Where it fits: Add these alongside or immediately after GFG Weeks 11–16. MIT 6.S191 is your reinforcement layer. Karpathy is your deep understanding layer.

MIT · Full Course · Updated Every Year (2026 Edition Now Live) · Free on YouTube

Course: https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5_7C0oIVt26ZgjG9NI

Lab: <https://introtodeeplearning.com/>

MIT's official deep learning course, updated annually and completely free. Unlike older courses, this covers modern topics — transformers, LLMs, diffusion models, RL — in a clean, structured way. Alexander Amini is an exceptional teacher. Every lecture has accompanying Colab notebooks you can run for free.

What you will learn:

- Deep learning foundations — neurons, backpropagation, optimisation
- CNNs — architecture, visualisation, transfer learning
- RNNs, Transformers, and Attention
- Deep Generative Modelling — VAEs, GANs, Diffusion
- Reinforcement Learning (dedicated lecture)
- Language Models and New Frontiers (dedicated lecture — covers LLMs)
- AI in biology, robotics, and science

5.6 — WQU Deep Learning Projects

→ Deep Learning Fundamentals Lab

<https://www.wqu.edu/deep-learning-lab>

5.7 a— Core Reinforcement Learning

5.7.1 Reinforcement Learning — *GFG Week 16*

5.7.2 Deep Learning — *Hugging Face Deep RL Course*

<https://huggingface.co/learn/deep-rl-course/unit0/introduction>

5.7.2 PPO (Proximal Policy Optimization) — *Hugging Face Deep RL Course*

<https://huggingface.co/learn/deep-rl-course/unit8/introduction>

5.7 b— David Silver: RL Course (UCL / DeepMind)

Where it fits: Do after GFG Week 16 and the Hugging Face Deep RL Course already in your roadmap. David Silver goes deeper into theory.

University College London + DeepMind · 10 Lectures · Free on YouTube

<https://www.youtube.com/playlist?list=PLqYmG7hTraZDM-OYHWgPebj2MfCFzFObQ>

David Silver invented AlphaGo. He is one of the top 3 RL researchers alive. This 10-lecture course is the mathematical foundation of modern RL — every paper in the field builds on the concepts taught here. It is older (2015) but the theory has not changed.

What you will learn:

- Markov Decision Processes — the mathematical framework for RL
- Dynamic Programming — value iteration, policy iteration
- Model-free prediction — Monte Carlo, TD learning
- Model-free control — SARSA, Q-learning
- Value function approximation — neural networks in RL
- Policy gradient methods — the foundation of PPO and TRPO
- Integrating learning and planning — Dyna, MCTS

- Exploration vs exploitation — the core RL tradeoff
- Case study: RL in classic games (foundation of AlphaGo)

5.8 — DSA

5.8.1 Problem Solving with Algorithms and Data Structures using Python — *Runestone Academy*

<https://runestone.academy/ns/books/published/pythonds/index.html>

5.8.2 DSA (Video lectures) — *freeCodeCamp*

<https://www.youtube.com/watch?v=pkYVOMU3MgA>

5.8.3 Practice Problems — *Striver's A2Z Sheet*

<https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z>

5.9 — Data Engineering

→ **Data Engineering Zoomcamp** — *DataTalksClub* · free · Kafka, Spark, dbt, BigQuery, Airflow

<https://datatalks.club/blog/data-engineering-zoomcamp.html>

PHASE 6 Modern AI Engineering



Companion Document

Phase 6 has a dedicated full course guide covering all topics below in depth, with the top 3 free comprehensive courses per subject, structured in the exact order you should study them.

→ **See: Modern AI Engineering — Full Course Guide**

Topics — in learning order

6.1 Natural Language Processing — *Transformers, BERT, GPT, text classification, NER, summarisation*

- 6.2 Computer Vision** — *CNNs, object detection, segmentation, OpenCV, real-time applications*
- 6.3 Large Language Models (LLMs)** — *Architecture, training dynamics, fine-tuning, deployment*
- 6.4 Generative AI** — *RLHF, diffusion models, full-stack GenAI engineering*
- 6.5 RAG — Retrieval-Augmented Generation** — *Building, evaluating, and optimising RAG pipelines*
- 6.6 Vector Databases** — *Embeddings, ANN search, ChromaDB, Weaviate, Pinecone, hybrid search*
- 6.7 Agentic AI** — *Agents, tool use, multi-agent systems, LangGraph, AutoGen*
- 6.8 Prompt Engineering + Fine-Tuning** — *Prompt design, LoRA, QLoRA, PEFT, instruction tuning*
- 6.9 FastAPI for AI Services** — *Building and deploying AI-powered APIs*

PHASE 7 Advanced Track — Cloud · Certification

📌 Complete Phases 5 and 6 before starting Phase 7. These topics require full deep learning and AI engineering knowledge to use meaningfully.

7.1 — Advanced MLOps at Scale

→ **MLOps Zoomcamp** — *DataTalksClub · complete if not done in Phase 4*
<https://datatalks.club/blog/mlops-zoomcamp.html>

7.2— Google Professional Machine Learning Engineer ★ Capstone

The Google Professional ML Engineer is one of the most respected cloud ML credentials globally. Attempt only after completing all phases above.

Step 1 Google Cloud Skills Boost — ML Path — *Official pre-certification curriculum — complete first*
<https://www.cloudskillsboost.google/paths/17>

Step 2 Official Exam Guide — *Free · everything testable is listed here*
<https://cloud.google.com/learn/certification/machine-learning-engineer>

Step 3 Preparing for the Professional ML Engineer Exam — *Google + Coursera · audit free · official exam prep*
<https://www.coursera.org/learn/preparing-cloud-professional-ml-engineer-exam>

Step 4 Hands-on Vertex AI Labs — *Google Cloud Skills Boost · complete every ML quest*
<https://www.cloudskillsboost.google/>

GeeksForGeeks Course (all GFG phases): <https://www.geeksforgeeks.org/batch/ds-16?tab=Resources>

DeepLearning.AI Short Courses (Phase 6 permanent resource): deeplearning.ai/short-courses

GPU Access — Kaggle Notebooks (Free T4 GPU · 30hrs/week · use for Phase 5, 6.8, and ARENA):
www.kaggle.com/code

GPU Access — Google Colab (Free T4 GPU backup · use when Kaggle quota runs out ·):
<https://colab.research.google.com/>